

1. Introduction

Bus service and passenger trip data can provide useful information about revenue, customer demand, route performance, occupancy, service delays, cancellations, customer satisfaction, and booking patterns. This project analyzes Cityflo-style bus service data across six major metro cities during 2024.

The raw dataset contains 3,258 trip records and 36 columns. It includes information about customers, cities, routes, bus types, trip dates, scheduled and actual timings, fares, discounts, payment modes, booking channels, trip status, cancellation reasons, ratings, occupancy, weather conditions, subscriptions, and complaints. The dataset was created synthetically for data-analysis practice and does not represent real Cityflo operational data.

The main goal of the project is to convert raw bus service data into useful business and operational insights. Python was used for data inspection, cleaning, exploratory analysis, and statistical testing, while Looker Studio was used to create the final interactive dashboard. The project development process follows a structured pipeline of data inspection, data cleaning, analysis, statistical testing, and dashboard reporting.

2. Project Objectives

The main objectives of this project are:

- 1. To understand overall trip, revenue, and customer performance.
- 2. To compare revenue and trip demand across cities and months.
- 3. To analyze trip completion, cancellation, and delay patterns.
- 4. To compare occupancy and service performance across different bus types.
- 5. To understand route demand and revenue performance.
- 6. To analyze customer ratings, complaints, age groups, and gender distribution.
- 7. To compare booking channels, subscription types, and payment modes.
- 8. To identify cancellation reasons and service reliability issues.
- 9. To understand weekday-wise trip demand across metro cities.
- 10. To present the findings through an interactive four-page bus service analytics dashboard.

3. Dataset Overview

The raw dataset contains 3,258 rows and 36 columns. The project notebook defines checks for data shape, column structure, duplicate records, extra spaces, missing values, incorrect data types, inconsistent category values, and invalid numerical values before analysis.

After cleaning and removing duplicate trip records, the final cleaned dataset contains 3,200 rows and 36 columns. A total of 58 duplicate records were removed during the cleaning process.

Important fields used in the analysis

Field	Description
trip_id	Unique trip identifier
customer_id	Unique customer identifier
gender	Customer gender
age_group	Customer age category
city	Metro city of operation
route_name	Bus route name
bus_type	Type of bus used for the trip

trip_date	Date of the trip
distance_km	Distance travelled in kilometres
fare_inr	Fare charged for the trip
discount_inr	Discount applied to the fare
net_revenue_inr	Revenue after discount
payment_mode	Payment method used by the customer
booking_channel	Channel used to book the trip
trip_status	Completed, Delayed-Completed, Cancelled, or No-show
cancellation_reason	Reason for trip cancellation
rating	Customer trip rating
occupancy_pct	Bus occupancy percentage
delay_minutes	Departure delay in minutes
subscription_type	Customer subscription category
complaint_raised	Indicates whether a complaint was raised
trip_month	Month of the trip
trip_weekday	Weekday of the trip

The cleaned dataset covers six metro cities: Mumbai, Pune, Bangalore, Hyderabad, Chennai, and Delhi NCR. The data also contains multiple routes and four bus categories: AC Seater, AC Sleeper, Non-AC Seater, and Premium AC.

4. Data Cleaning and Preparation

Before analysis, the dataset was checked and prepared to make the data reliable for reporting.

The main preparation steps included:

- Checking the structure and size of the dataset.
- Checking duplicate records.
- Checking missing values.
- Removing extra spaces from text fields.
- Standardizing categorical values.
- Standardizing inconsistent gender values.
- Converting boolean-style fields into a common True/False format.
- Converting fare values into the correct numerical format.
- Checking and correcting invalid age and occupancy values.
- Converting trip dates into the correct date format.
- Removing unnecessary columns.
- Creating additional analytical fields.
- Checking cancellation and trip-status consistency.
- Removing duplicate trip records.
- Exporting the prepared dataset for further analysis.

The raw data contained inconsistent text and data formats. For example, extra spaces were found in customer name, gender, and city fields, while `fare_inr` was stored as text and `trip_date` was not initially stored as a date field. Boolean fields such as `is_peak_hour`, `gps_enabled`, and `complaint_raised` also contained different representations that required standardization.

Several columns that were not required for the final analysis were removed, including booking ID, customer name, driver ID, driver name, bus number, seat number, and device type.

Additional fields were then created for the analysis, including trip year, trip month, trip weekday, delay minutes, delayed-trip flag, net revenue, and age group.

The cleaning process also checked logical consistency between trip status and cancellation information. After duplicate removal and final validation, the cleaned dataset contains 3,200 records and 36 columns.

5. Key Performance Indicators

The dashboard uses a common KPI section to provide a quick summary of the workforce.

Key Performance Indicator	Dashboard Value	Formula / Definition	Business Significance
Net Revenue	₹859,623.50	SUM(net_revenue_inr)	Shows the total revenue generated after discounts.
Total Trips	3,200	COUNT_DISTINCT(trip_id)	Shows the total number of trips covered by the analysis.
Unique Customers	1,251	COUNT_DISTINCT(customer_id)	Shows the total number of different customers.
Completed Trips	2,494	Completed + Delayed-Completed	Shows the number of trips that were successfully completed.
Cancellation Rate	22.06%	Cancelled + No-show / Total Trips	Shows the share of trips that were not completed.
On-Time Performance	68.75%	Trips with delay_minutes ≤ 5 / Total Trips	Shows the overall level of service punctuality.
Avg Occupancy	57.86%	AVG(occupancy_pct)	Shows the average percentage of bus capacity used.
Avg Rating	3.80	AVG(rating)	Shows the average customer rating.
Complaint Rate	6.41%	Complaint Records / Total Trips	Shows the share of trips where a complaint was raised.

The project notebook also identifies revenue, trips, completed trips, cancellation rate, on-time performance, average occupancy, average rating, complaint rate, and city-level revenue as important dashboard metrics.

The final analysis reports ₹859,623.50 in net revenue, 3,200 total trips, 2,494 completed trips, a 22.06% cancellation rate, 68.75% on-time performance, 57.86% average occupancy, 3.80 average rating, and a 6.41% complaint rate.

6. Main Calculated Metrics

Several calculated metrics were used to make the dashboard more meaningful.

6.1 Net Revenue

Net Revenue = Fare – Discount

This metric represents the actual revenue generated from a trip after subtracting the discount amount. The net_revenue_inr field was created from fare_inr and discount_inr.

6.2 Completed Trips

Completed Trips = Completed + Delayed-Completed Trips

This metric counts all trips that were successfully completed, including trips completed after a delay.

6.3 Cancellation Rate

$$\text{Cancellation Rate} = (\text{Cancelled} + \text{No-show Trips}) / \text{Total Trips}$$

This metric shows the proportion of trips that were not successfully completed.

6.4 On-Time Performance

$$\text{On-Time Performance} = \text{Trips with Delay} \leq 5 \text{ Minutes} / \text{Total Trips}$$

This metric shows the share of trips operating within the accepted five-minute delay limit. The dashboard value is **68.75%**.

6.5 Average Occupancy

$$\text{Average Occupancy} = \text{AVG}(\text{occupancy_pct})$$

This metric shows the average percentage of bus capacity used across trips. The dashboard value is **57.86%**.

6.6 Average Rating

$$\text{Average Rating} = \text{AVG}(\text{rating})$$

This metric shows the average customer rating for trips where a rating is available. The dashboard value is **3.80 out of 5**.

6.7 Average Late Arrival and Early Exit

$$\text{Complaint Rate} = \text{Complaint Records} / \text{Total Trip Records}$$

This metric shows the proportion of trip records where a customer complaint was raised. The dashboard value is **6.41%**.

7. Analytical and Statistical Findings

The analysis included descriptive analysis and statistical testing. Four main statistical tests were performed to study the relationship between trip distance and fare, peak-hour travel and delay, bus type and fare, and city and trip status. The results help identify which differences or relationships in the dataset are statistically meaningful.

7.1 Distance and Fare

Pearson correlation was used to study the relationship between `distance_km` and `fare_inr`. The purpose of this test was to determine whether trip fare changes systematically with travel distance. The notebook defines the null hypothesis as no linear correlation between distance and fare and the alternative hypothesis as the presence of a linear relationship.

However, the final notebook output returned Pearson **r = NaN** and **p-value = NaN**. Therefore, this test did not produce a valid correlation coefficient or significance value, and no reliable statistical conclusion should be drawn from this particular result.

7.2 Total Hours Worked and Productive Hours

An independent two-sample t-test was used to compare average delay minutes between peak-hour and non-peak-hour trips.

The analysis reported an average delay of **5.4 minutes for peak-hour trips** and **5.1 minutes for non-peak-hour trips**. The test produced **t = 0.550** and **p = 0.5824**.

Since the p-value is greater than 0.05, the analysis found **no statistically significant difference in average delay between peak-hour and non-peak-hour trips**. This suggests that peak-hour travel alone did not have a significant effect on trip delay in this dataset.

7.3 Bus Type and Fare

A one-way ANOVA was used to compare average fare across the four bus types.

The analysis found a statistically significant difference in average fare between bus types, with $F = 86.53$ and $p = 1.46 \times 10^{-53}$.

The average fares were:

- Premium AC — ₹379.3
- AC Sleeper — ₹343.8
- AC Seater — ₹276.0
- Non-AC Seater — ₹272.9

Premium AC recorded the highest average fare, while Non-AC Seater recorded the lowest. The statistical result indicates that bus type has a significant relationship with fare level in the dataset.

7.4 City and Trip Status

A Chi-Square test of independence was used to study whether trip status is associated with city.

The analysis produced **Chi-square = 12.80, degrees of freedom = 15, and $p = 0.6181$** . Since the p-value is greater than 0.05, the test found **no statistically significant relationship between city and trip status**.

This suggests that the distribution of **Completed, Delayed-Completed, Cancelled, and No-show trips** was broadly similar across the six metro cities. Although individual city counts differ, the differences are not statistically strong enough to conclude that trip-status behavior depends on city.

8. Looker Studio Dashboard

The final dashboard is divided into four pages, with each page focusing on a different area of Cityflo bus service performance.

The dashboard contains common filters for **Date Range, City, Bus Type, and Trip Status**. The same KPI cards are also displayed across all four pages to provide a consistent summary of overall performance.

The four dashboard pages are:

1. Executive Overview
2. Revenue & Demand Analysis
3. Operations & Service Performance
4. Customer & Booking Insights

8.1 Page 1 — Executive Overview

Purpose :

This page provides a high-level overview of Cityflo business and trip performance. It focuses on monthly revenue and trip volume, city-level revenue, overall trip status, and demand across bus types.

8.1.1 Monthly Revenue & Trip Volume

Chart Type: Combo Chart

Dimension: Month

Metric: Net Revenue and Total Trips

This chart compares monthly net revenue with total trip volume. Net Revenue is displayed using bars, while Total Trips are displayed using a line. It helps understand whether changes in trip demand are also reflected in monthly revenue.

The monthly results are:

- January — ₹71,997 from 262 trips
- February — ₹64,868 from 242 trips
- March — ₹79,246 from 299 trips
- April — ₹80,376 from 278 trips
- May — ₹67,226 from 255 trips
- June — ₹65,697 from 252 trips
- July — ₹71,855 from 279 trips
- August — ₹74,277 from 275 trips
- September — ₹65,692 from 251 trips
- October — ₹72,557 from 265 trips
- November — ₹79,351 from 296 trips
- December — ₹66,485 from 246 trips

April generated the highest monthly net revenue at approximately ₹80.4K (278 trips), while February generated the lowest at approximately ₹64.9K (242 trips). March recorded the highest trip volume with 299 trips (₹79.2K), followed by November with 296 trips (₹79.4K). February had the lowest trip volume with 242 trips (₹64.9K).

8.1.2 Net Revenue by City

Chart Type: Bar Chart

Dimension: City

Metric: SUM(net_revenue_inr)

This chart compares total net revenue generated across the six metro cities. It helps identify the cities contributing the most and least to overall revenue.

The city-wise revenue values are:

- Mumbai — ₹157,032
- Pune — ₹153,194
- Bangalore — ₹148,379
- Chennai — ₹140,450
- Hyderabad — ₹135,077
- Delhi NCR — ₹125,493

Mumbai generated the highest revenue, followed closely by Pune and Bangalore. Delhi NCR recorded the lowest revenue among the six cities.

The difference between the highest and lowest city revenue is approximately ₹31.5K, showing that revenue is distributed reasonably well across cities rather than being concentrated in only one location.

8.1.3 Trip Status Distribution

Chart Type: Donut Chart

Dimension: trip_status

Metric: Trip Count

This chart shows the overall distribution of Completed, Delayed-Completed, Cancelled, and No-show trips.

The final trip-status distribution is:

- Completed — 2,111 trips (66%)
- Delayed-Completed — 383 trips (12%)
- Cancelled — 373 trips (11.7%)
- No-show — 333 trips (10.4%)

Completed trips form the largest category and represent nearly two-thirds of all trip records. When Completed and Delayed-Completed trips are combined, the dashboard records 2,494 successfully completed trips.

Cancelled and No-show trips together account for 706 trips, which produces the overall 22.06% cancellation rate displayed in the KPI section.

8.1.4 Trip Volume by Bus Type

Chart Type: Treemap

Dimension: bus_type

Metric: Trip Count

This chart compares total trip volume across the four bus types and helps identify which bus categories are used most frequently.

The trip counts are:

- Non-AC Seater — 826 trips
- AC Sleeper — 801 trips
- Premium AC — 787 trips
- AC Seater — 786 trips

Non-AC Seater recorded the highest trip volume, while AC Seater recorded the lowest. However, the difference between all four bus categories is relatively small.

This indicates that trip demand is distributed quite evenly across the available bus types, with no single bus type dominating overall usage.

8.2 Page 2 — Revenue & Demand Analysis

Purpose :

This page focuses on revenue trends and customer demand. It compares monthly revenue, route-level demand and revenue, booking channels, subscription types, and city-wise weekday trip demand.

8.2.1 Monthly Net Revenue Trend

Chart Type: Time Series Chart

Dimension: Month

Metric: SUM(net_revenue_inr)

This chart shows how net revenue changes across the twelve months of 2024.

Revenue increased from approximately ₹72K in January to ₹79.2K in March and reached its annual highest level of ₹80.4K in April. Revenue then decreased during May and June before improving again during July and August.

September recorded another decline to approximately ₹65.7K, followed by recovery in October and November. November generated approximately ₹79.4K, making it the second-highest revenue month. Revenue declined again to approximately ₹66.5K in December.

The monthly trend shows that revenue fluctuates throughout the year rather than following a continuous upward or downward pattern. March, April, and November were the strongest revenue months, while February, June, September, and December were comparatively weaker.

8.2.2 Route Demand vs Net Revenue

Chart Type: Scatter Chart

Dimension: route_name

X-Axis Metric: Total Trips

Y-Axis Metric: Net Revenue

Trend Line: Linear

This chart compares trip demand with net revenue across individual routes. Each point represents one route, while the trend line shows the overall relationship between route demand and revenue.

Route demand ranges from approximately 34 to 69 trips, while route-level revenue ranges from approximately ₹4.1K to ₹30.1K.

The Kharadi - Magarpatta route generated the highest route-level revenue at approximately ₹30,084 from 62 trips. The route with the highest trip count recorded 69 trips, but did not generate the highest revenue.

The upward trend line suggests that routes with more trips generally tend to generate higher revenue. However, the points are widely spread, showing that trip volume alone does not fully explain revenue. Fare level, route distance, bus type, and discount patterns can also affect route revenue.

8.2.3 Net Revenue by Booking Channel

Chart Type: Vertical Bar Chart

Dimension: booking_channel

Metric: SUM(net_revenue_inr)

This chart compares the total revenue generated through each booking channel.

The revenue values are:

- Website — ₹225,059.5
- Corporate Portal — ₹224,631
- Mobile App — ₹207,995
- Kiosk — ₹201,938

Website generated the highest revenue, closely followed by Corporate Portal. The difference between these two channels is very small.

Mobile App and Kiosk generated slightly lower revenue, with Kiosk recording the lowest contribution. Overall, all four booking channels make meaningful contributions to total revenue, which suggests that Cityflo demand is distributed across multiple booking platforms.

8.2.4 Revenue Contribution by Subscription Type

Chart Type: Pie Chart

Dimension: subscription_type

Metric: SUM(net_revenue_inr)

This chart shows the share of total net revenue generated by each subscription type.

The revenue contribution is:

- Monthly Pass — ₹220,160 (25.6%)
- Weekly Pass — ₹214,891.5 (25%)
- Corporate Plan — ₹214,738 (25%)
- Single Ride — ₹209,834 (24.4%)

Monthly Pass generated the highest revenue contribution, while Single Ride generated the lowest.

However, all four subscription types contribute almost equally to revenue. Each category contributes approximately one-quarter of total revenue, which indicates a well-balanced revenue mix rather than dependence on one specific subscription category.

8.2.5 Trip Demand by City & Weekday

Chart Type: Pivot Table With Heatmap

Rows: City

Columns: Weekday

Metric: Total Trips

This heatmap compares trip demand across cities and weekdays. Darker or stronger cells represent relatively higher trip volumes.

Important demand patterns include:

- Pune on Tuesday — 106 trips, the highest city-weekday combination.
- Mumbai on Monday — 97 trips
- Pune on Friday — 92 trips
- Hyderabad on Tuesday — 92 trips
- Delhi NCR on Wednesday — 89 trips
- Pune on Thursday — 89 trips

The lowest demand was recorded for Pune on Saturday with 55 trips, followed by Hyderabad on Monday with 57 trips.

The heatmap shows that weekday demand is not identical across cities. For example, Pune shows particularly strong Tuesday and Friday demand, while Mumbai performs strongly on Monday and Friday.

8.3 Page 3 — Operations & Service Performance

Purpose :

This page focuses on operational reliability and service quality. It analyzes average delays across cities, trip-status composition, bus-type occupancy and delay, completed-trip delay categories, and reasons for cancellation.

8.3.1 Average Delay by City

Chart Type: Vertical Bar Chart

Dimension: City

Metric: AVG(delay_minutes)

Filter: Completed and Delayed-Completed Trips

This chart compares the average delay across the six cities for successfully completed trips.

The city-wise average delay values are:

- Hyderabad — 6.68 minutes
- Delhi NCR — 6.47 minutes
- Chennai — 6.21 minutes
- Mumbai — 5.44 minutes
- Pune — 5.4 minutes
- Bangalore — 5.3 minutes

Hyderabad recorded the highest average delay, followed by Delhi NCR and Chennai. Bangalore recorded the lowest average delay.

The difference between Hyderabad and Bangalore is approximately 1.38 minutes, indicating some variation in punctuality across cities but not an extremely large gap.

8.3.2 Trip Status by City

Chart Type: 100% Stacked Bar Chart

Dimension: City

Breakdown Dimension: trip_status

Metric: Trip Count

This chart compares the composition of Completed, Delayed-Completed, Cancelled, and No-show trips across cities.

The city-level counts are:

- Mumbai: 378 Completed, 63 Delayed-Completed, 57 Cancelled, 55 No-show
- Pune: 369 Completed, 58 Delayed-Completed, 60 Cancelled, 65 No-show
- Delhi NCR: 344 Completed, 71 Delayed-Completed, 74 Cancelled, 48 No-show
- Bangalore: 351 Completed, 58 Delayed-Completed, 64 Cancelled, 62 No-show
- Chennai: 339 Completed, 61 Delayed-Completed, 65 Cancelled, 52 No-show
- Hyderabad: 330 Completed, 72 Delayed-Completed, 53 Cancelled, 51 No-show

Mumbai recorded the highest number of Completed trips, while Hyderabad recorded the lowest Completed count.

Delhi NCR had the highest number of Cancelled trips, while Pune recorded the highest number of No-show trips.

Despite these differences, the overall trip-status composition remains fairly similar across cities, which is also consistent with the Chi-Square analysis reported earlier.

8.3.3 Bus Type Occupancy & Delay

Chart Type: Combo Chart

Dimension: bus_type

Metrics: Average Occupancy and Average Delay (min)

Filter: Completed and Delayed-Completed Trips

This chart compares bus capacity utilization with service delay across different bus types.

The results are:

- AC Sleeper — 59.12% Avg Occupancy, 5.5 min Avg Delay
- Non-AC Seater — 58.01% Avg Occupancy, 6.43 min Avg Delay
- AC Seater — 57.22% Avg Occupancy, 6.04 min Avg Delay
- Premium AC — 56.22% Avg Occupancy, 5.64 min Avg Delay

AC Sleeper recorded the highest average occupancy and relatively low delay, making it strong from both capacity-utilization and service-reliability perspectives.

Premium AC recorded the lowest average occupancy, while Non-AC Seater recorded the highest average delay.

The chart shows that higher occupancy does not necessarily result in higher delay. AC Sleeper, for example, has the highest occupancy but a lower average delay than Non-AC Seater and AC Seater. The current dashboard shows:

8.3.4 Trip Delay Status

Chart Type: Donut Chart

Dimension: Delay Category

Metric: Trip Count

Filter: Completed and Delayed-Completed Trips

This chart divides the 2,494 completed trips into different delay categories.

The results are:

- On Time (0–5 min) — 1,867 trips (74.9%)
- Minor Delay (6–15 min) — 293 trips (11.7%)
- Severe Delay (30+ min) — 198 trips (7.9%)
- Moderate Delay (16–30 min) — 136 trips (5.5%)

Nearly three-quarters of completed trips were on time within the five-minute threshold.

Minor delays represent the second-largest category, while moderate delays represent the smallest share. Severe delays account for approximately 8% of completed trips and therefore remain an important area for operational improvement.

The percentage in this chart is higher than the overall On-Time Performance KPI because this chart analyzes completed trips only, while the KPI is calculated using the broader trip dataset.

8.3.5 Cancellation Reasons

Chart Type: Treemap

Dimension: cancellation_reason

Metric: Trip Count

This chart shows the main reasons associated with the 706 Cancelled and No-show trips.

The cancellation reasons are:

- Customer Request — 306 trips
- Payment Failed — 163 trips
- Bus Breakdown — 72 trips
- Traffic/Weather — 65 trips
- Driver Unavailable — 57 trips
- Route Diverted — 43 trips

Customer Request is clearly the largest reason, accounting for more than two-fifths of all cancellation-related records.

Payment Failed is the second-largest reason and represents nearly one-quarter of cancellation cases. Operational reasons such as bus breakdown, traffic/weather, driver availability, and route diversion individually contribute smaller shares.

8.4 Page 4 — Customer & Booking Insights

Purpose :

This page focuses on passenger characteristics, customer satisfaction, complaint patterns, payment preferences, and booking behavior.

8.4.1 Average Delay by City

Chart Type: 100% Stacked Vertical Bar Chart

Dimension: age_group

Breakdown Dimension: gender

Metric: Trip Count

This chart compares gender composition across customer age groups.

The gender distribution is:

1. 18–25 Age Group
 - Female — 52.59%
 - Male — 44.03%
 - Other — 3.39%
2. 26–35 Age Group
 - Female — 49.82%
 - Male — 45.46%
 - Other — 4.71%
3. 36–45 Age Group
 - Female — 51.5%
 - Male — 42.68%
 - Other — 5.83%
4. 46–60 Age Group
 - Female — 38.27%
 - Male — 54.32%
 - Other — 7.41%

Female passengers form the largest share in the 18–25, 26–35, and 36–45 age groups. In the 46–60 group, male passengers form the largest share.

The Other category remains the smallest segment across all age groups, although its share increases slightly in the older age groups.

8.4.2 Customer Rating Distribution

Chart Type: Vertical Bar Chart

Dimension: Rating

Metric: Rated Trips

This chart shows how customer ratings are distributed from 1 star to 5 stars.

The valid rating counts are:

- 1 Star — 108
- 2 Stars — 195
- 3 Stars — 469
- 4 Stars — 618
- 5 Stars — 755

A total of 2,145 valid ratings are available.

Five-star ratings form the largest category, followed by four-star ratings. Together, 4-star and 5-star ratings account for approximately 64% of all valid ratings.

Low ratings are much less common, with 1-star and 2-star ratings together accounting for approximately 14% of valid responses. This supports the overall dashboard average rating of 3.80 out of 5.

8.4.3 Customer Experience by City

Chart Type: Bubble Chart

Dimension: City

X-Axis Metric: Average Rating

Y-Axis Metric: Complaint Rate

Bubble Size: Unique Customers

This chart compares customer satisfaction and complaint behavior across cities.

The city-level values are approximately:

- Mumbai — Avg Rating 3.88, Complaint Rate 5.8%, 553 Unique Customers
- Delhi NCR — Avg Rating 3.86, Complaint Rate 6.9%, 536 Unique Customers
- Hyderabad — Avg Rating 3.79, Complaint Rate 6.7%, 506 Unique Customers
- Bangalore — Avg Rating 3.76, Complaint Rate 6.9%, 535 Unique Customers
- Pune — Avg Rating 3.75, Complaint Rate 6.3%, 552 Unique Customers
- Chennai — Avg Rating 3.76, Complaint Rate 5.8%, 517 Unique Customers

Mumbai shows the strongest overall customer-experience combination, with the highest average rating and one of the lowest complaint rates.

Bangalore has one of the lower average ratings and the highest complaint rate, while Pune has the lowest average rating but the largest unique-customer count.

The chart helps identify cities where customer satisfaction is relatively strong as well as locations where complaint reduction may require greater attention.

8.4.4 Payment Mode Usage

Chart Type: Pie Chart

Dimension: payment_mode

Metric: Trip Count

This chart shows the distribution of payment methods among records where payment information is available.

The payment-mode counts are:

- UPI — 492 (17.4%)
- Wallet — 484 (17.1%)
- Debit Card — 470 (16.6%)
- Cash — 466 (16.5%)
- Net Banking — 458 (16.2%)
- Credit Card — 457 (16.2%)

UPI is the most frequently used payment method, while Credit Card has the smallest count.

However, the difference across payment methods is very small. Each payment mode contributes around 16%–17% of valid payment records, showing highly balanced customer payment preferences.

8.4.5 Payment Modes by Booking Channel

Chart Type: Pivot Table With Heatmap

Rows: booking_channel

Columns: payment_mode

Metric: Trip Count

This chart shows the main reasons associated with the 706 Cancelled and No-show trips.

This chart compares payment-method usage across different booking channels.

The detailed values are:

1. Website — Total 739
 - Cash — 129
 - Credit Card — 120
 - Debit Card — 118
 - Net Banking — 107

- UPI — 134
- Wallet — 131
- 2. Corporate Portal — Total 719
 - Cash — 114
 - Credit Card — 115
 - Debit Card — 101
 - Net Banking — 124
 - UPI — 128
 - Wallet — 137
- 3. Mobile App — Total 701
 - Cash — 113
 - Credit Card — 117
 - Debit Card — 125
 - Net Banking — 121
 - UPI — 125
 - Wallet — 100
- 4. Kiosk — Total 668
 - Cash — 110
 - Credit Card — 105
 - Debit Card — 126
 - Net Banking — 106
 - UPI — 105
 - Wallet — 116

Website recorded the highest number of valid payment records, followed by Corporate Portal, Mobile App, and Kiosk.

The highest individual booking-channel/payment combination is Corporate Portal with Wallet at 137 records, followed by Website with UPI at 134 records.

Overall, the heatmap shows that payment methods are distributed relatively evenly within each booking channel. No single payment method completely dominates any one channel.

9. Key Dashboard Insights

The four-page dashboard provides a combined view of Cityflo's revenue, trip demand, operational performance, and customer behavior. The main insights are:

1. Overall trip completion is strong, with 2,494 out of 3,200 trips completed successfully. However, the 22.06% cancellation rate shows that cancellations and no-shows still represent an important service issue.
2. Cityflo generated a total net revenue of ₹859,623.50 during 2024. April recorded the highest monthly revenue at ₹80,376, while February recorded the lowest at ₹64,867.50.
3. Mumbai generated the highest city-level revenue at ₹157,031.50, followed by Pune and Bangalore. Delhi NCR recorded the lowest revenue at ₹125,492.50.
4. Monthly revenue and trip volume generally move in a similar direction, but higher trip volume does not always produce the highest revenue. This indicates that revenue is also influenced by factors such as route distance, fare, bus type, and discounts.
5. Revenue contribution is well distributed across booking channels. Website and Corporate Portal are the strongest channels, generating approximately ₹225K each, while Mobile App and Kiosk also make substantial contributions.
6. Revenue is also highly balanced across subscription types. Monthly Pass contributes the highest share at about 25.6%, but Weekly Pass, Corporate Plan, and Single Ride each contribute close to one-quarter of total revenue.
7. Trip demand varies across cities and weekdays. Pune on Tuesday recorded the highest city-weekday demand with 106 trips, while Pune on Saturday recorded the lowest with 55 trips.
8. Operational performance shows that most completed trips operate within acceptable delay levels. Around 74.86% of completed trips were on time within five minutes, while severe delays accounted for 7.94%.
9. Hyderabad recorded the highest average delay, while Bangalore recorded the lowest. Bus-type analysis also shows that higher occupancy does not necessarily result in greater delay.
10. Customer Request is the largest cancellation reason, accounting for 306 cases or about 43.3% of cancellation-related trips. Payment Failed is the second-largest reason with 163 cases.

11. Customer satisfaction is generally positive. The dashboard reports an average rating of 3.80 out of 5, and approximately 64% of valid ratings are either 4-star or 5-star.
12. Customer payment preferences are highly balanced. UPI is the most commonly used payment mode, but Cash, Wallet, Debit Card, Credit Card, and Net Banking all have similar usage levels.

Overall, the dashboard shows a fairly balanced business across cities, booking channels, subscription types, bus types, and payment methods, while cancellations, service delays, and customer-request cancellations remain the main areas for improvement.

10. Business Recommendations

10.1 Reduce Trip Cancellations

The overall cancellation rate is 22.06%, which shows that cancellations and no-shows are still an important service issue. Customer Request is the largest reason, accounting for 43.3% of cancellation-related trips, followed by Payment Failed at 23.1%.

Cityflo should improve booking confirmation, reminder messages, easy rescheduling options, and payment reliability. These steps may help reduce avoidable cancellations and improve successful trip completion.

10.2 Improve Service Punctuality

The overall On-Time Performance is 68.75%, while 74.86% of completed trips fall within the 0–5 minute delay range. However, some completed trips still experience moderate and severe delays.

Cities with relatively higher average delays, especially Hyderabad and Delhi NCR, should be reviewed for route scheduling, traffic conditions, dispatch timing, and other operational issues. Since the statistical test found no significant difference between peak-hour and non-peak-hour delays, punctuality improvement should focus on overall operations rather than only peak-hour services.

10.3 Optimize Route and Bus Allocation

Trip demand varies across cities and weekdays. For example, Pune on Tuesday recorded the highest demand with 106 trips, while Pune on Saturday recorded only 55 trips.

Bus frequency and capacity can therefore be adjusted according to actual demand. Higher-demand routes and days may require additional capacity, while lower-demand periods can be reviewed to avoid underutilization. The overall average occupancy of 57.86% also shows that there is room to improve bus-capacity utilization.

10.4 Strengthen Revenue and Booking Strategy

Website and Corporate Portal are the strongest booking channels, each generating approximately ₹225K in revenue. Mobile App and Kiosk also make important contributions, although their revenue is slightly lower.

Revenue is also well balanced across Monthly Pass, Weekly Pass, Corporate Plan, and Single Ride subscriptions, with each contributing about one-quarter of total revenue. Cityflo should continue supporting multiple booking and subscription options while improving engagement through relatively weaker channels.

10.5 Improve Customer Experience

The overall average customer rating is 3.80 out of 5, and around 64% of valid ratings are 4-star or 5-star, which indicates generally positive customer feedback.

However, the complaint rate is 6.41%, so customer complaints, low ratings, cancellations, and delay issues should be reviewed together. Cities or routes showing repeated service problems can be monitored more closely so that customer satisfaction and service quality can be improved over time.

Since the statistical analysis found no significant relationship between city and trip status, common service standards can generally be maintained across cities while specific local operational issues are handled separately.

11. Limitations

The analysis has a few limitations:

- The dataset is synthetically generated for EDA practice and is not real Cityflo operator data, so the findings represent simulated service patterns rather than actual business performance.
- The dataset covers trips from 2024 only, so longer-term trends across multiple years cannot be evaluated.
- Some fields contain context-dependent missing values, such as ratings or trip timing information for cancelled trips.
- Statistical significance does not automatically prove a direct cause-and-effect relationship between the variables.
- The Pearson correlation test between distance and fare returned NaN values, so no reliable statistical conclusion was drawn from that particular test.
- Some dashboard measures use specific filters, such as completed-trip filters for delay analysis, so their values may differ from overall KPIs calculated across all trip records.

12. Conclusion

This project converts raw Cityflo bus-service data into a structured analytics solution for understanding revenue, trip demand, operational performance, and customer behavior. The analysis combines data cleaning, exploratory analysis, calculated metrics, statistical testing, and interactive visualization using Python and Looker Studio.

The final four-page dashboard provides a clear view of Executive Overview, Revenue & Demand Analysis, Operations & Service Performance, and Customer & Booking Insights. It allows users to compare cities, bus types, booking channels, subscription types, trip status, delays, occupancy, ratings, complaints, and payment behavior using consistent measures.

Overall, the project demonstrates how structured data analysis and dashboard visualization can turn trip-level transport data into useful insights for revenue monitoring, demand planning, service improvement, operational decision-making, and customer-experience analysis. Since the dataset is synthetically generated, the results represent simulated Cityflo-style service patterns rather than actual operator performance.